



Smart Reporting System to Assist Relevant Authorities in Improving Emergency Response

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ABSTRACT

Traditional crime and emergency reporting systems face significant challenges in keeping pace with digital transformation, often relying on manual methods and voice communications. This reliance results in delayed response times, inaccurate geographic data, and inefficient management of security resources. This project aims to develop a "Smart Reporting System" to assist relevant authorities in improving emergency response by providing a unified digital platform consisting of a mobile application for citizens and a centralized web dashboard for security authorities.

The system utilizes Artificial Intelligence (AI) and deep learning algorithms to automatically classify reports and analyze spatial patterns to generate Heat Maps, identifying high-risk areas to support proactive decision-making. It leverages Global Positioning System (GPS) technology to ensure precise incident location and employs Know Your Customer (KYC) and face recognition technologies to verify user identity, thereby ensuring report credibility and reducing false alarms. The system is developed using React Native for the mobile application, Python/FastAPI for backend processing, and PostgreSQL with PostGIS for spatial data management. The anticipated results demonstrate that the system contributes to reducing response times, enhancing transparency between citizens and security agencies, and establishing a centralized database that supports sustainable security planning.

Keywords:

Smart Reporting System, Emergency Response, Artificial Intelligence (AI), Heat Maps, Geolocation (GPS), Deep Learning, Crime Classification.

الملخص

تواجه أنظمة الإبلاغ التقليدية عن الجرائم والحالات الطارئة تحديات كبيرة في مواكبة التحول الرقمي، حيث تعتمد غالباً على الاتصالات الصوتية والأساليب اليدوية، مما يؤدي إلى تأخر الاستجابة، ونقص دقة البيانات الجغرافية، وصعوبة في إدارة الموارد الأمنية. يهدف هذا المشروع إلى تطوير "نظام إبلاغ ذكي" لمساعدة الجهات المختصة في تحسين الاستجابة للحالات الطارئة، من خلال توفير منصة رقمية موحدة تعتمد على تطبيق هاتف محمول للمواطنين ولوحة تحكم مركزية للسلطات الأمنية.

يوظف النظام تقنيات الذكاء الاصطناعي (AI) وخوارزميات التعلم العميق لتصنيف البلاغات تلقائياً وتحليل الأنماط المكانية لإنشاء خرائط حرارية (Heat Maps) تحدد المناطق الأكثر عرضة للمخاطر، مما يدعم اتخاذ القرارات الاستباقية. كما يعتمد النظام على تقنيات تحديد الموقع الجغرافي (GPS) لضمان دقة مكان الحادث، ويستخدم تقنيات التحقق من الهوية (KYC) والتعرف على الوجه لضمان مصداقية البلاغات وتقليل البلاغات الكاذبة. تم بناء النظام باستخدام إطار عمل React Native لتطبيقات الهاتف، و Python/FastAPI للمعالجة الخلفية، وقاعدة بيانات PostgreSQL مع PostGIS للبيانات المكانية. تظهر النتائج المتوقعة أن النظام يساهم في تقليل زمن الاستجابة، تعزيز الشفافية بين المواطن والأمن، وتوفير قاعدة بيانات مركزية تدعم التخطيط الأمني المستدام.

الكلمات المفتاحية: نظام الإبلاغ الذكي، الاستجابة للطوارئ، الذكاء الاصطناعي، الخرائط الحرارية، تحديد الموقع الجغرافي (نظام تحديد المواقع العالمي)، التعلم العميق، تصنيف الجرائم.

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ABBRAVIATIONS

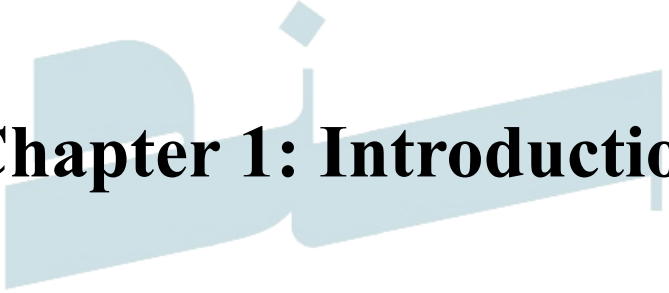
Acronym	Definition
CSS	Cascading Style Sheets
DOM	Document Object Model
GIS	Geographic Information System
GPS	Global Positioning System
HTML	Hypertext Markup Language
iOS	iPhone Operating System
IT	Information Technology
JS	JavaScript
KYC	Know Your Customer
LTE	Long-Term Evolution
OCR	Optical Character Recognition
PHP	Hypertext Preprocessor
RDBMS	Relational Database Management System
REST	Representational State Transfer
SQL	Structured Query Language
TLS	Transport Layer Security
UI	User Interface
YOLO	You Only Look Once

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Chapter 1: Introduction

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1.1. Introduction

Accidents and emergencies of all kinds, such as crimes, traffic accidents, and fires, are among the major challenges facing modern societies, given their direct impact on individual safety and social stability. With increasing population density and the accelerating pace of life, traditional methods relying on voice communication or in-person intervention are no longer sufficient to meet the demands for speed and accuracy in reporting and response.

The concept of security and public safety is no longer limited to responding to incidents after they occur; it now depends on the speed of reporting, the accuracy of information, and the ability to analyze data to support timely decision-making. Delayed reporting or a lack of geographical information weakens the efficiency of the relevant authorities and increases the potential extent of damage.

"Despite the improvement in emergency management with the emergence of modern communication methods, reliance on traditional reports and non-automated routing mechanisms is no longer sufficient in light of current challenges, which calls for the adoption of smarter, more efficient solutions based on modern technologies." [1]

More than half the world's population now lives in urban areas, and in most countries, citizens use smartphones. Smartphones have become the primary means of communication and computing for daily tasks.[2] This has led to the need for intelligent electronic systems for reporting incidents and emergencies, capable of providing accurate geolocation, sending reports instantly and securely, and ensuring data confidentiality. Furthermore, advancements in artificial intelligence technologies have enhanced the

ability of these systems to analyze data and identify patterns and high-risk areas.

Contemporary research confirms that integrating artificial intelligence technologies into incident reporting systems enables improved classification accuracy and information analysis, providing predictive models that can be used to identify high-risk areas and respond faster and more efficiently.[3]

Based on this, this project aims to design and develop an intelligent electronic system for reporting accidents and emergencies of all kinds, which contributes to improving response efficiency, supporting analysis and planning processes, and enhancing trust between the community and the competent authorities, in line with the requirements of digital transformation and enhancing public safety.

1.2. Background

Emergency and crime reporting systems have deep roots in human history, adapting and evolving alongside the increasing complexity of urban patterns and technological innovations. In ancient times, societies relied on rudimentary, instinctive methods of warning, such as smoke, fire, or auditory signals like drumming and loud calls. With the rise of large cities in the 17th century, more organized public warning systems emerged, employing bells in central squares to alert residents and authorities to incidents.

The mid-19th century witnessed a significant development in reporting systems with the emergence of telegraph-based alarm systems, followed by the development of wired electrical alarm systems later in the century. With

the dawn of the 20th century, standardized emergency numbers emerged, enabling citizens to communicate directly with the relevant authorities. The first dedicated emergency system was launched in London in 1937, followed by the 911 number in the United States in 1968. These systems contributed to improved public safety and security for extended periods, but they remained primarily reliant on voice communication and direct human intervention, which limited their ability to provide immediate and accurate response in complex urban environments. [4]

With the dawn of the third millennium, the digital revolution and the development of Geographic Information Systems (GIS) and Global Positioning Systems (GPS) contributed to the shift towards intelligent systems that rely on real-time data and precise location tracking. This development helped to reduce the information gaps that plagued traditional systems, which in some urban areas led to significant deficiencies in security data.

Relying on traditional methods in emergency management leads to a critical delay in decision-making due to the mental strain on human operators; therefore, integrating artificial intelligence systems as an intermediary helps reduce the time to process a report by more than 40%. [5]

In recent years, artificial intelligence and data analytics have become prominent elements in the development of emergency decision support systems, where data analytics algorithms are used to process large amounts of information and improve the understanding of critical situations in real time. Recent studies show that integrating artificial intelligence can help in developing decision support systems and analyzing data patterns related to emergencies, thus supporting response and preparedness strategies. [6]

Despite this progress, many reporting systems still face challenges related to data integration, information accuracy, and the reliability of reports, especially in environments requiring rapid response and high-level coordination among multiple entities. This underscores the need to develop more integrated and intelligent electronic systems capable of supporting public safety requirements and keeping pace with the digital transformation of emergency services.

1.3. Problem Statement

The research problem lies in the inability of traditional crime reporting systems to keep pace with rapid digital transformations and the increasing security needs of contemporary urban environments. These systems still rely heavily on manual methods and direct voice communication between citizens and operations rooms. This reliance leads to poor accuracy of the information received, particularly regarding the geographic location of reports, which negatively impacts response times and often delays security intervention at critical moments.

These systems also suffer from a clear deficiency in managing and directing field response units due to the absence of real-time tracking mechanisms and a lack of an accurate operational picture of the locations of patrols and specialized units. This makes it difficult to assign the nearest or most readily available unit to handle reports, leading to the misuse of limited security resources and hindering the ability of competent authorities to evaluate and improve response efficiency.

Furthermore, the current system largely relies on paper documentation or simple, disjointed digital systems, which impedes the development of a

comprehensive, centralized database that can be relied upon for analyzing crime patterns, supporting data-driven security planning, and predicting future risks. This deficiency has contributed to a significant information gap, reducing the effectiveness of security decision-making and fostering a reactive rather than proactive approach to crime prevention.

At the societal level, this technological weakness and lack of digital transparency have led to a decline in trust between citizens and security agencies, given the absence of mechanisms that allow citizens to track the status of their reports or monitor their processing. Therefore, there is a clear need to develop an intelligent electronic crime reporting system that utilizes location-based technologies and artificial intelligence. This system should aim to improve the efficiency of security responses, enhance coordination among relevant authorities, and rebuild public trust through a unified digital platform that is fast and reliable.

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1.4. The Importance of the system

The system's importance lies in its role as a strategic tool for enhancing security and overcoming the physical barriers and risks associated with traditional reporting processes. It allows citizens to document crimes safely, addressing the concerns of 78% of the population who fear traveling to police stations during conflict. Furthermore, it contributes to improved human resource management, which has seen a 55% decrease in personnel, by automating the processing of routine reports and freeing up staff for more complex tasks.

In addition, the system bridges a 70% information gap by building a centralized digital database that supports the analysis of crime hotspots and the proactive allocation of resources based on precise criminal patterns. Institutionally, the project aims to rebuild public trust, which has declined to 35%, through real-time tracking features that enhance transparency. With 75% smartphone penetration and 65% internet coverage, the system represents a highly efficient technological solution that guarantees up to 87% improvement in security response times.

1.5. Main Objectives

1. Empowering users and enhancing digital trust by providing a secure and user-friendly environment for submitting and tracking reports while ensuring data privacy.
2. Achieving operational efficiency in report management through automated classification and intelligent assignment to relevant authorities, thereby reducing response times.

3. Supporting proactive planning and security decision-making by analyzing the spatial and temporal patterns of reports.

1.6. Sub-objectives

1. Enabling users to track the status of reports in real time from submission to closure.
2. Reducing the percentage of false or inaccurate reports through automated verification tools.
3. Building a reliable database that can be analyzed and retrieved.
4. Creating heat maps that show the areas most prone to incidents.
5. Supporting decision-makers with analytical reports and smart dashboards.

1.7. Project Motivation

The motivations for this project stem from the need to address the security and technical challenges facing conflict-affected cities, particularly the shortcomings of traditional reporting systems and the weak response of relevant authorities. Based on academic studies and reports from international organizations, the project's motivations can be summarized as follows:

1. The inefficiency of traditional reporting systems in post-conflict environments, where reliance on manual methods and voice communication leads to difficulties in resource management, crime pattern analysis, and decision-making.

2. Limited access for citizens to security agencies. International reports indicate that fear of movement and geographical barriers constitute a major obstacle to reporting crimes in areas with unstable security.
3. An information gap resulting from weak documentation of security data. This limits the authorities' ability to conduct data-driven security planning and accurate spatial analysis.
4. Pressure on limited security resources and personnel in crisis-affected environments. Automating the receipt and classification of reports is essential to improve response efficiency and resource allocation.
5. Declining public trust in security institutions. Studies confirm that the lack of transparency and follow-up mechanisms negatively impacts reporting rates and cooperation between the community and relevant authorities.
6. A relatively suitable technological environment for adopting digital solutions. This is due to the increasing prevalence of smartphones and internet services, even in conflict-affected areas, which supports the shift towards electronic reporting systems.
7. The global trend is towards smart systems that rely on artificial intelligence and geographic information systems (GIS) to improve crisis management, crime analysis, and enhance community security.

1.8. Project Idea

The project revolves around developing a smart security platform based on a mobile application powered by artificial intelligence. This platform will serve as an instant communication channel between citizens and relevant security authorities. The system allows citizens to securely report incidents through a user-friendly interface that supports the transmission of geographic data and multimedia, thereby improving the quality of security documentation and expediting the reporting process.

The system also acts as a decision support tool, processing and classifying reports and analyzing spatial data to create heat maps. These maps help authorities identify priority security areas and support proactive resource allocation planning, while maintaining the human element in decision-making. Furthermore, the system provides real-time geographic information, suggested routes, and estimated arrival times, enabling efficient deployment of response units. It also offers a transparent tracking mechanism that fosters trust between citizens and security institutions.

1.9. Project Features

1. **High Geographic Accuracy:** The system provides precise location pinpointing of reports using GPS technology, addressing the gap in verbal descriptions found in traditional systems and ensuring faster response times for response teams.
2. **Intelligent and Proactive Analysis:** The system integrates an artificial intelligence module that automatically categorizes reports and generates heat maps (hotspots) to identify crime-prone areas, supporting proactive security decision-making.

3. **Integrated Digital Documentation:** The application supports the uploading of multimedia, including photos, videos, and audio recordings, providing a solid digital evidence base that enhances the quality of security investigations.
4. **Real-Time Synchronization and Rapid Response:** The system boasts high performance, ensuring that reports are sent and unit locations are updated in no more than 3 seconds, with routing capabilities to identify the shortest geographical routes.
5. **Transparency and Report Tracking:** The system allows citizens to track the status of their reports in real time (sent, processed, completed), contributing to bridging the trust gap between the community and security agencies .
6. **Superior Security and Protection:** The system employs advanced encryption standards (AES-256) to protect stored data and TLS 1.3 protocols for transmitted data, along with a robust authorization system that prevents record tampering.
7. **Flexibility and Scalability:** The system is designed with a cloud architecture that ensures service continuity and the ability to handle massive loads of up to 100,000 concurrent users in extreme emergencies.

1.10. Project Scope

The scope defines the technical and operational framework of the system, outlining the technical boundaries and target use categories to ensure efficient security management.

1.10.1. Technical Scope

The system's technical architecture is based on a distributed cloud architecture that includes the following components:

1. **Mobile Applications (React Native):** Development of a unified cross-platform application using the React Native framework to ensure seamless user interfaces and superior performance for citizens and response units on both Android and iOS systems.
2. **Centralized Management Platform:** Building a dedicated web dashboard for security authorities to monitor reports and analyze data in real time.
3. **Infrastructure and Artificial Intelligence:** Creating a geographic database (PostGIS) and integrating machine learning (AI) models to classify crimes and analyze risk hotspots.
4. **Digital Integration Services:** Relying on Google Maps APIs and push notification services to ensure real-time synchronization.

1.10.2. Scope Of Use

The system is designed as a flexible model applicable in the following environments:

1. **Individuals and whistleblowers:** a secure platform for submitting media-documented reports and transparently tracking their processing status.
2. **Security operations rooms:** a central tool for decision support and task allocation based on intelligent data analysis.
3. **Field emergency teams:** coordinating the movement of police and ambulance units and providing them with the shortest geographical routes.
4. **Sustainable urban security:** as part of smart city infrastructure to enhance stability and protect vital facilities and residential areas.

1.11. Project Limitations

1. **Technical Reliability:** The overall system efficiency depends on the availability of continuous internet coverage (WiFi/4G/LTE) and a stable power supply for charging phones.
2. **Limited Initial Data:** The accuracy of AI algorithms may be low in the initial stages of operation until a large database of models is built.
3. **False Reports:** The security challenge of misuse of the application through the submission of false reports necessitates additional verification mechanisms.
4. **Privacy and Legality:** Strict legal restrictions govern the handling and encryption of criminal data and sensitive personal media.
5. **Digital Divide:** The system may be difficult to adopt for those lacking sufficient technical skills or modern smartphones.

1.12. Methodology

This project is based on the applied and experimental approach (the mixed approach), where this approach relies on collecting and analyzing data to reach accurate results, and focuses on developing scientific solutions to a specific problem and proceeds as follows:

1. **Needs Analysis and Data Collection:** The current reporting systems in urban environments are studied to determine the technical requirements for citizens and security agencies. This phase includes collecting geographic, multimedia, and historical crime data to ensure the system's accurate database is built.
2. **System Development:** Theoretical concepts are translated into a tangible software tool using the React Native framework to develop a mobile application. The application is linked to a geographic database and an integrated web management platform to ensure seamless data flow between citizens and the operations room.
3. **Software Implementation and AI Support:** This step involves programming AI algorithms and integrating them with the application interfaces. This aims to automate the processing and real-time classification of reports and analyze geographic patterns to generate heat maps.
4. **AI Testing and Verification:** The system undergoes pilot testing in a simulated environment to measure the AI algorithms' ability to accurately classify reports. A confusion matrix is used to calculate the accuracy and reliability of the results.

5. **Response Time Testing and Performance Evaluation:** The time elapsed from the moment a report is submitted until its arrival and dispatch in the field is measured. The application's stability is also tested under the pressure of simultaneous reports, and GPS accuracy is examined under various conditions.
6. **Results Analysis and Recommendations:** This phase begins by comparing the experimental results with the planned objectives to ensure the system's effectiveness in improving security. The process concludes with the formulation of technical and operational recommendations for future system development based on the collected data.

1.13. Summary

This chapter presents the foundational framework for the Smart Crime Reporting System project, starting with an analysis of the shortcomings of traditional methods and culminating in the desired objectives of the modern system. It also reviews the importance and motivations behind developing the project, the system's technical features, and provides a brief explanation of the AI-based concept and the hybrid methodology used to implement and test the system to ensure its operational effectiveness.

Chapter 2: Theoretical framework

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2.1 Overview

This chapter addresses the theoretical principles and scientific concepts that form the backbone of intelligent reporting systems, highlighting the evolutionary path of these systems and their transition from limited traditional methods to advanced digital environments. It presents a comprehensive view of the architecture of modern reporting systems, with a focus on the role of artificial intelligence as a core element in processing, analyzing, and automatically classifying data, thereby contributing to more effective prioritization of emergency cases.

Furthermore, the chapter reviews the technical standards related to data security and digital identity verification, emphasizing their importance in enhancing system reliability and building trust between users and concerned authorities. In addition, it examines the geographic and software technologies used to link real-world field data with cloud-based systems, illustrating how these tools integrate to create a real-time response framework capable of overcoming geographical and temporal challenges. This chapter represents an essential knowledge foundation for understanding the mechanisms that ensure operational efficiency in modern emergency environments.

2.2. Reporting Systems

2.2.1. Traditional Reporting Systems

Traditional reporting systems rely on manual methods such as paper reports or direct telephone calls, where reports are recorded and processed manually by designated staff. Despite their simplicity, these systems suffer from several problems, most notably slow response times, frequent human error, difficulty in data storage and retrieval, and weak follow-up and documentation mechanisms. Furthermore, the heavy reliance on human

intervention leads to decreased operational efficiency, especially as the number of reports increases.

2.2.2. Electronic Reporting Systems

With technological advancements and digital transformation, electronic reporting systems have emerged, relying on web and smartphone applications to receive reports. These systems allow users to easily submit reports at any time, with the option to attach photos and geographic location. This type of system has contributed to faster and more efficient report reception compared to traditional systems. However, some electronic systems still suffer from limitations in intelligent data analysis and the inability to accurately and automatically classify reports.

2.2.3. Smart Reporting Systems

It uses artificial intelligence to analyze incoming data in real time, providing high accuracy in geolocation and automated incident classification.

AI-powered emergency response systems have emerged as critical enablers for improving real-time decision-making, risk assessment, and disaster mitigation across various sectors, enhancing prediction accuracy and reducing intervention delays.[7]

Real-time incident detection powered by machine learning enables systems to identify emergencies more quickly, reduce false alarms, and allocate resources more efficiently.[7]

2.3.importance of reporting systems

Reporting systems play a pivotal role in improving service delivery across various sectors, particularly in security and public services. They facilitate

communication between users and relevant authorities, expedite response times, and mitigate potential risks. These systems also aid in data collection and analysis to generate indicators and statistics that support decision-making and enhance the efficiency of planning and monitoring.

2.3.1. Comparison between electronic reporting systems and intelligent reporting systems

Table 1 Comparison between electronic reporting and intelligent reporting systems

<i>Comparison Criteria</i>	<i>Electronic Reporting Systems (without AI)</i>	<i>Intelligent Reporting Systems (with AI)</i>
<i>Report Receiving Mechanism</i>	Direct electronic receipt	Intelligent electronic receipt with report content analysis
<i>Report Classification</i>	Manual classification or according to fixed rules	Automatic classification using AI algorithms
<i>Processing Speed</i>	Moderate and relies on partial human intervention	Very high due to intelligent automation
<i>Routing Accuracy</i>	Errors may occur in routing reports	Accurate routing to the competent authority based on intelligent analysis
<i>Data Handling</i>	Data storage and display only	Data analysis and pattern and indicator extraction
<i>Learning and Development Capability</i>	Fixed and does not improve with use	Capable of learning and improving over time
<i>Decision Support</i>	Limited and relies on traditional reports	Strong thanks to predictive and proactive analysis
<i>Handling Large Volumes of Reports</i>	Efficiency decreases with increasing number of reports	Maintains efficiency even with large volumes

2.4. Identity verification

Identity verification is a fundamental element of modern digital systems, particularly in applications that handle sensitive data or offer security-related services. This process ensures that the user is the legitimate and authorized individual using the system, thereby enhancing the reliability of the services provided and mitigating identity theft or misuse of the platform.[8]

The Know Your Customer (KYC) system is the first line of defense in ensuring the credibility of incoming reports. Its objective is to link the digital account to a legitimate, legal identity, preventing the creation of fake accounts and reducing frivolous reports that drain emergency resources.

2.4.1. Know you customer (KYC)

KYC is considered a radical solution to the problem of random reporting that plagues traditional systems. This is achieved through "identity verification" to ensure that every user is a real person who can be legally held accountable. [9]

This process relies on three integrated technical pillars: it begins with the automated extraction of identity data

1. Optical Character Recognition (OCR):

Optical Character Recognition (OCR) is a technology that uses automated data extraction to quickly convert images of text into a machine-readable format. [10]

It is used to extract data from official identity documents (such as national identity cards or passports). The system works by processing the image of the identity document captured by the application, converting the printed text into readable and analyzable digital data (such as name, national identification number, and date of birth). This helps reduce human error during registration and ensures that the entered data matches the attached official documents.



Figure 1 OCR MECHANISM

2. Face Recognition:

Facial recognition is a method of identifying or confirming an individual's identity using an image of their face. Facial recognition

systems can be used to identify people in real time or to identify people in photos or videos.[11]

In the context of Know Your Customer (KYC), facial recognition plays a crucial role in digital identity verification. The process begins with face matching, where a user's selfie is compared to the photograph taken from an official identification document (such as a national identity card). This is intended to ensure that the person opening the account is indeed the document holder, thus reducing the risk of fraud and identity theft.[12]

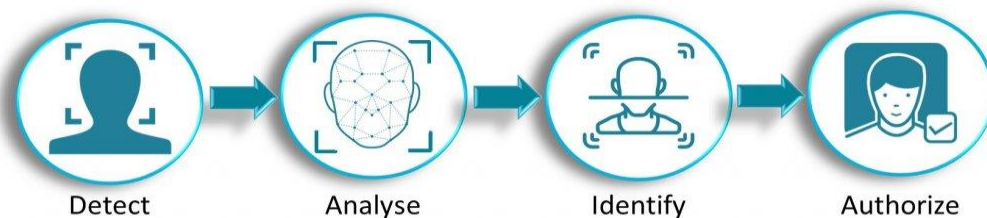


Figure 2 Face Recognition

3. Liveness Detection:

Biometric verification, or biometric testing, is a security technique that ensures an entity providing biometric data, such as a face or fingerprint, is a living person and not a fake image or recording.[13]

This technique represents the most important layer of security to prevent spoofing. It aims to confirm that the face in front of the camera is a real, living being and not just a printed image, recorded video, or 3D mask. This is typically done through interactive tests (active liveness tests) such as asking for a blink or a smile to detect any attempt at manipulation before the recording process is completed.

2.5. Artificial intelligence

Artificial intelligence (AI) is a broad branch of computer science that aims to build systems capable of mimicking human cognitive abilities. AI is not limited to automating routine tasks; it extends to include the ability to learn, reason, solve problems, and perceive the surrounding environment. In research, it is considered the overarching umbrella under which machine learning and deep learning technologies fall. In projects, the goal is to enable the system to make intelligent decisions based on input data without explicitly programming it for each individual case.



Figure 3 ARTIFICIAL INTELLIGENCE

2.6. AI Agent

An AI Agent is an autonomous system designed to execute tasks by constructing workflows using available tools. The functional scope of AI agents extends far beyond natural language processing, encompassing decision-making, problem-solving, interaction with external environments, and the execution of complex actions. [15]

In this system, the AI Agent acts as the "mastermind." It receives data outputs from computer vision models (such as YOLO) and integrates them into the broader context of the situation. Instead of merely sending a notification for a "fire," the agent analyzes surrounding variables—such as geographic location, weather conditions, and the proximity of emergency units—to make a logical decision regarding the severity level and the most appropriate authority for response. This approach minimizes the cognitive load on human operators and ensures that every report is handled based on a holistic view of the situation rather than an isolated image.

2.7.Computer Vision

Computer vision is a branch of artificial intelligence that equips machines with the ability to process, analyze, and interpret visual inputs, such as images and videos. This technology uses machine learning to help computers and other systems extract meaningful information from visual data[14].

Computer vision is the cornerstone of a system's ability to perceive and understand surrounding events, acting as a sophisticated "eye" that automatically and instantly detects information. Instead of relying solely on potentially incomplete or inaccurate human descriptions in the moment of danger, computer vision analyzes incoming visual footage (images and videos) to accurately determine the nature of the incident. Whether it's detecting a fire by identifying smoke and flames, recognizing traffic accidents, or even monitoring suspicious behavior, this technology ensures that emergency teams have reliable and detailed visual information that enables them to make informed decisions.

The function of computer vision in this project extends beyond simply recognizing objects, encompassing two main phases:

2.7.1. Digital Forgery Detection

The system verifies the validity of the report before processing it. This is done through two technical pathways within the system that analyze subtle visual patterns to detect any manipulation. Through the Moire Pattern Detection model, optical interferences resulting from screen re-enactments are monitored to ensure the recency of the event, while the EfficientNet model detects

deepfake images generated by artificial intelligence or modified by design software, thus ensuring that only genuine and reliable reports reach the authorities.

2.7.2. Accident analysis

Incidents are analyzed and prioritized in real time. Using advanced algorithms such as *YOLOv8-Seg* and *YOLOv8-Detect*, the system not only detects the presence of a fire but also estimates its size and spread using Instance Segmentation, providing a comprehensive view of the risk level. Furthermore, the integration of technologies like *Temporal Analysis* and *ByteTrack* allows the system to monitor the course and development of an incident (such as tracking the movement of people within an incident or the spread of a fire), enabling the AI agent to make immediate and accurate decisions about the most appropriate response and the need for reinforcements.

2.8. Technologies used in development

2.8.1. Internet Technology

The internet is defined as an electronic communications network that connects millions of computers worldwide. It comprises millions of governmental, academic, commercial, and small government networks. The internet is simultaneously a global broadcasting medium, a mechanism for disseminating information, and a means of collaboration and interaction between individuals and their computers, regardless of their geographical location. Today, the internet is replete with a vast amount of data and services, perhaps the most widely used of which are web pages and mobile applications.

2.8.2.Web Technologies

Intelligent reporting systems rely on modern web technologies to build their back-end and front-end user interfaces, enabling the creation of interactive and user-friendly platforms. Application programming interfaces (APIs) are also used to connect the various system components and ensure secure and efficient data exchange between users and servers.

2.8.3.Application Technologies

Smartphone application technologies play a pivotal role in the proposed smart reporting system, serving as the primary interface through which users interact with the system. The mobile application allows users to easily submit reports at any time, with the ability to attach photos or videos and accurately pinpoint their geographic location.

Modern mobile applications rely on development frameworks that enable the creation of interactive and user-friendly interfaces, while ensuring high performance and rapid response. These technologies also provide direct integration with the system's back-end server via application programming interfaces (APIs), enabling secure and immediate data transfer.

2.8.4.Deep Learning

It is an advanced subfield of machine learning that relies on artificial neural networks (ANNs) for its structure.

How it works: Deep learning mimics the way the human brain works by using multiple layers of artificial neurons (called hidden layers).

Its importance: Its strength lies in its superior ability to handle unstructured data such as images, text, and audio. It automatically extracts features from the data without requiring significant manual human intervention, which distinguishes it from traditional machine learning.

2.8.5.Geo-Location

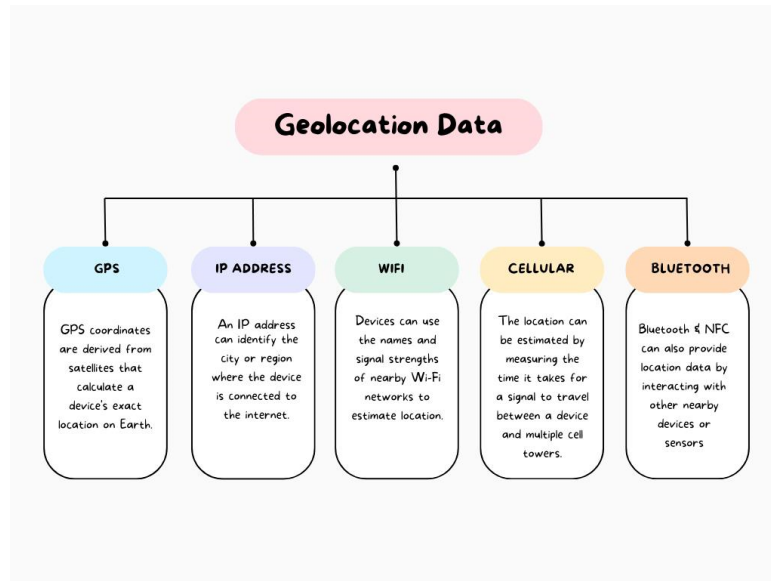


Figure 4 Geo-Location

Geolocation technology is defined as the technical procedure used to determine the geographic coordinates (latitude and longitude) of a network-connected digital device in real time. In the smartphone environment, this technology relies on a hybrid approach that integrates data from multiple sources to ensure accuracy.

Geolocation technology operates through a series of integrated technical processes that begin with the physical environment and end with the user interface.

This mechanism can be explained in the following sequence:

1. **Sensing and Signal Collection Phase:** The process begins when the smartphone's antenna searches for available signals in the surrounding environment. It primarily relies on radio waves from satellites within the GPS system. If the user is in enclosed or densely populated urban areas, the device collects supplementary data from nearby Wi-Fi network IDs and cellular towers to expedite the initial search area.

2. **Mathematical Processing and Triangulation:** Once a minimum signal is detected, the operating system performs complex calculations known as triangulation. This process measures the time it takes for the signal to travel from the source to the phone and then calculates the precise distance between them. By intersecting these distances from multiple sources, the point of convergence, representing the device's precise geographic location, is determined as a set of digital coordinates consisting of longitude and latitude.

3. Reverse Geocoding

Since digital coordinates are incomprehensible to the average user or security personnel, these numbers are transmitted over the internet to specialized map servers (such as Google Maps). These servers perform a process called "reverse geocoding" to convert these coordinates into textual and descriptive data, including the country, city, neighborhood, and even the name of a nearby street. This facilitates the accurate reading and location of the security report on the map.

4. Data Integration and Analysis in the System

In the final stage of implementing crime reporting, the documented geographic location is automatically integrated with other report details, such as the time and recorded images. This information package is sent to a central database, where it appears immediately to receiving agencies. This collected geographic data then allows artificial intelligence algorithms to analyze spatial patterns of crime and create heat maps that illustrate the distribution of criminal activity in specific geographic areas, thus supporting data-driven security decision-making.

2.8.6.Front-end Technologies

2.8.6.1. Mobile Application

React Native : React Native is a powerful framework for developing cross-platform smartphone applications. It enables the creation of Android and iOS apps using a single JavaScript codebase and a component-based approach that accelerates development and facilitates code reuse, making it suitable for projects requiring multiple applications. It also delivers near-native performance by relying on actual system components and supports leveraging smart device features such as GPS, camera, and push notifications—essential elements for accurate reporting and rapid response in emergencies.



Figure 5 Front-end Technologies

2.8.6.2. Web Dashboard

1. HTML

"Hypertext Markup Language, or HTML, is the standard markup language for describing the structure of documents displayed on the web. HTML consists of a series of elements and attributes that are used to label all the components of a document to organize it in a meaningful way." [10]

2. CSS

Cascading Style Sheets (CSS) provides the overall look, or presentation, of a page. CSS is used to define how HTML documents are displayed. With CSS, we can set the page layout, font properties, colors, sizes, borders, backgrounds, and the positioning of elements on the page. It is a markup language like HTML, and the two are often used together. CSS enables the display to be adjusted to suit different types of devices, such as large and small screens or printers. CSS is independent of HTML and performs some



Figure 6 CSS Language

functions that HTML cannot, making it useful for scaling content across different platforms, such as laptops and mobile phones.

3. React JS Framework

React JS was used to develop the web interface for the authorities' control panel due to its advanced capabilities in building interactive and dynamic interfaces that rely on real-time data updates, with enhanced performance thanks to Virtual DOM technology. React also allows the interface to be organized into independent components, making it suitable for complex administrative systems and helping relevant authorities efficiently track reports, maps, and statistics, and make quick decisions based on real-time data updates.



Figure 7 React JS Framework

2.8.7.Back-end Technologies

Python:

was chosen for the system's backend development due to its simplicity, power, and extensive support for building interfaces and managing databases. It is used for processing mobile application requests, the control panel, and implementing the report management logic. Python also forms the foundation for the system's artificial intelligence module, thanks to its advanced libraries for data analysis and machine learning. This enables report classification, geographic pattern analysis, and the creation of heat maps to support decision-makers and enhance security response efficiency.



Figure 8 Back-end Technology

FastAPI:

The FastAPI framework was adopted for developing the system's RESTful APIs due to its high performance and Python-based architecture. FastAPI is characterized by its responsiveness and support for asynchronous programming, making it suitable for handling real-time requests from mobile applications and the control panel.

FastAPI provides automatic API documentation using the Swagger UI, simplifying service testing and integration with front-ends. It also supports data validation and permissions management mechanisms, enhancing system security and ensuring the integrity of data exchange between its various components.



Figure 9 Back-end Technology

2.8.8.Databases

PostgreSQL:

PostgreSQL is an advanced, open-source relational database management system (RDBMS) known for its high reliability and robust support for handling complex data. For this project, PostgreSQL was chosen to store user data, reports, responses, and activity logs because of its advanced support for constraints, transactions, and referential integrity mechanisms, ensuring data safety and accuracy.



Figure 10 Data base Technologies

PostGIS:

PostGIS is a spatial extension of the PostgreSQL database, adding advanced capabilities for handling geographic and spatial data. PostGIS enables the storage of geographic coordinates for reports, location analysis, and the calculation of distances between the report location and response units—a crucial component of a system based on GPS.

PostGIS supports spatial analysis functions such as identifying hotspots, analyzing the geographic distribution of crime, and displaying the results on interactive maps within the authorities' dashboard. This represents a significant technical foundation for employing artificial intelligence in predicting high-risk areas and optimizing the allocation of security resources.

2.8.9. GitHub

GitHub is an essential tool for code management and version control throughout project development. It enables collaborative work among development team members by tracking changes, managing branches, and integrating code in a structured and secure manner.

GitHub also contributes to project quality improvement through change documentation, the ability to revert to previous versions, and support for code review mechanisms. This is particularly important for sensitive systems projects, such as security systems, as these tools reduce bugs and increase the reliability of the final system.



Figure 11 Git TECHNOLOGY

2.9.Previous studies

Table 2 the related works to our supposed project

STUDY TITLE	STUDY AUTHOR	STUDY DATE	STUDY FLAWS
Crime record management system	BIJAY KUMAR SHAH 20IT103012	2022	1. THE SYSTEM IS A WEB APPLICATION AND DOES NOT PROVIDE A MOBILE APP FOR CITIZENS TO REPORT PROBLEMS IMMEDIATELY. 2. THE SYSTEM RELIES ON STORING AND DISPLAYING REPORTS TO MANAGEMENT; IT DOES NOT UTILIZE ARTIFICIAL INTELLIGENCE. 3. ALL REPORTS ARE TREATED EQUALLY; THERE IS NO EVALUATION OR PRIORITIZATION PROCESS. 4. THE STUDY RELIED ON TRADITIONAL WEB TECHNOLOGIES SUCH AS PHP AND MYSQL.
Web-based citizen-police report system	Yee Suzan, Nayef Abdulwahab Mohammed Alduais*	2023	1. MANUAL REVIEW OF DRAINS BY OFFICIALS. 2. NO ARTIFICIAL INTELLIGENCE TECHNOLOGIES WERE USED. 3. NO DATA PROTECTION METHOD (ENCRYPTION) WAS MENTIONED.
Design and implementation of online crime reporting system for the Nigeria police force	Major Awwal ISA (Ph.D.) ¹	2024	1. THE SYSTEM FUNCTIONS AS AN ARCHIVING TOOL, RECEIVING AND STORING DATA AS IS. OUR SYSTEM, HOWEVER, EMPLOYS INTELLIGENT ANALYSIS, UTILIZING ARTIFICIAL INTELLIGENCE TO AUTOMATICALLY CLASSIFY REPORTS. 2. IMAGES AND FILES ARE UPLOADED ONLY AS ATTACHMENTS; THERE IS NO ANALYSIS OF THE CONTENT OF IMAGES AND VIDEOS. 3. THERE IS NO VERIFICATION OF USER IDENTITY; REGISTRATION IS DONE WITH A USERNAME AND PASSWORD.

			4. ANY USER CAN UPLOAD A REPORT; THERE IS NO INTELLIGENT VERIFICATION OF THE REPORT'S CREDIBILITY.
Kollona Amn	Saudi Ministry of Interior	2016	1. RELIES ON MANUAL INPUT AT ALL STEPS AND MANUAL SELECTION OF THE RESPONSE UNIT. 2. THERE IS NO INTELLIGENT ANALYSIS OF THE CONTENT OF IMAGES AND VIDEOS. 3. THE APPLICATION IS MERELY A MEANS OF TRANSMITTING REPORTS.

2.10. Conclusion

This chapter reviewed the theoretical and technical foundations of intelligent reporting systems, clarifying their evolution from traditional methods to digital systems that rely on artificial intelligence and spatial analysis. The chapter demonstrated the role of these technologies in improving report classification, accelerating emergency response, and enhancing the accuracy of decision-making. It also addressed the importance of data security and digital identity verification in ensuring system reliability and building trust between users and relevant authorities. This chapter constitutes a fundamental knowledge base that paves the way for analyzing the current system and designing the proposed solution in subsequent chapters.



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